

```
package com.mycompany.proyectofinalv1;
```

```
import interfaz.*;
```

```
import javax.swing.SwingUtilities;
```

```
/**
```

```
 *
```

```
 * @author sam10
```

```
 */
```

```
public class ProyectoFinalV1 {
```

```
    public static void main(String[] args) {
```

```
        SwingUtilities.invokeLater(() -> {
```

```
            ApplicationContext context = new ApplicationContext();
```

```
            new LoginFrame(context).setVisible(true);
```

```
        });
```

```
    }
```

```
}
```

```
package interfaces;
```

```
import java.time.LocalDateTime;
```

```
import java.util.ArrayList;
```

```
import modelo.Asistencia;
```

```
import modelo.Usuario;
```

```
/**
```

```
 *
```

```

* @author sam10
*/

public interface IAsistenciaServicio {

    void registrarAsistencia(Asistencia asistencia);

    boolean registrarSalida(Usuario usuario, LocalDateTime fechaHoraSalida);

    ArrayList<Asistencia> obtenerReporteAsis(Usuario usuario);
}

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to
change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to edit
this template
 */

package interfaz;

import modelo.*;
import javax.swing.*;
import java.awt.*;

/**
 *
 * @author sam10
 */

public class LoginFrame extends javax.swing.JFrame {

```

```

private ApplicationContext context;

private JTextField txtId;

private JPasswordField txtPassword;

private JLabel lblMensaje;


private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(LoginFrame.class.getName());


/**
 * Creates new form LoginFrame
 */
public LoginFrame(ApplicationContext context) {

    this.context = context;

    initComponents();

    setLocationRelativeTo(null);

    this.getContentPane().setBackground(new java.awt.Color(230, 240, 255));

    this.setResizable(false);

    getRootPane().setDefaultButton(btnLogin);
}


/**
 * This method is called from within the constructor to initialize the form.
 * WARNING: Do NOT modify this code. The content of this method is always
 * regenerated by the Form Editor.
 */
@SuppressWarnings("unchecked")

```

```
// <editor-fold defaultstate="collapsed" desc="Generated Code">

private void initComponents() {

    jScrollBar1 = new javax.swing.JScrollBar();
    lblTitulo = new javax.swing.JLabel();
    lblId = new javax.swing.JLabel();
    lblPassword = new javax.swing.JLabel();
    idText = new javax.swing.JTextField();
    passwordText = new javax.swing.JPasswordField();
    btnLogin = new javax.swing.JButton();
    jLabel4 = new javax.swing.JLabel();
    mensajeLbl = new javax.swing.JLabel();

    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
    setTitle("Coordinacion médica");
    setBackground(new java.awt.Color(0, 204, 204));
    setResizable(false);

    lblTitulo.setText("Coordinación médica");

    lblId.setText("ID");

    lblPassword.setText("Contraseña");

    idText.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
```



```

        .addGroup(layout.createSequentialGroup())

        .addGap(94, 94, 94)

    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

        .addComponent(lblPassword,
javax.swing.GroupLayout.PREFERRED_SIZE, 69,
javax.swing.GroupLayout.PREFERRED_SIZE)

        .addComponent(lblId, javax.swing.GroupLayout.PREFERRED_SIZE,
23, javax.swing.GroupLayout.PREFERRED_SIZE))

    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)

    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

        .addComponent(idText)

        .addComponent(passwordText))))

    .addGroup(layout.createSequentialGroup())

    .addGap(179, 179, 179)

    .addComponent(jLabel4, javax.swing.GroupLayout.PREFERRED_SIZE, 37,
javax.swing.GroupLayout.PREFERRED_SIZE))

    .addGroup(layout.createSequentialGroup())

    .addGap(149, 149, 149)

    .addComponent(btnLogin))

    .addGroup(layout.createSequentialGroup())

    .addGap(110, 110, 110)

    .addComponent(mensajeLbl, javax.swing.GroupLayout.PREFERRED_SIZE,
173, javax.swing.GroupLayout.PREFERRED_SIZE))

    .addContainerGap(117, Short.MAX_VALUE))

);

layout.setVerticalGroup(

```

```

layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(layout.createSequentialGroup()
        .addGap(47, 47, 47)
        .addComponent(lblTitulo)
        .addGap(57, 57, 57)

    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(lblId)
        .addComponent(idText, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(20, 20, 20)

    .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
        .addComponent(lblPassword)
        .addComponent(passwordText, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(35, 35, 35)
        .addComponent(btnLogin)
        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
        .addComponent(jLabel4)

    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
        .addComponent(mensajeLbl, javax.swing.GroupLayout.PREFERRED_SIZE, 19,
javax.swing.GroupLayout.PREFERRED_SIZE)
        .addContainerGap(21, Short.MAX_VALUE))

```

```
);
```

```
pack();
```

```
// </editor-fold>
```

```
private void idTextActionPerformed(java.awt.event.ActionEvent evt) {
```

```
    // TODO add your handling code here:
```

```
}
```

```
private void btnLoginActionPerformed(java.awt.event.ActionEvent evt) {
```

```
    // TODO add your handling code here:
```

```
String id = idText.getText().trim();
```

```
String pass = new String(passwordText.getPassword());
```

```
Usuario u = context.iniciarSesion(id, pass);
```

```
if (u == null) {
```

```
    mensajeLbl.setText("ID o contraseña incorrectos");
```

```
    return;
```

```
}
```

```
if (u instanceof Admin admin) {
```

```
    AdminFrame af = new AdminFrame(context, admin);
```

```
    af.setVisible(true);
```

```
    dispose();
```



```

    }

    else if (u instanceof Estudiante est) {

        EstudianteFrame ef = new EstudianteFrame(context, est);

        ef.setVisible(true);

        dispose();

    }

    else {

        JOptionPane.showMessageDialog(this, "Tipo de usuario no soportado",

            "Error", JOptionPane.ERROR_MESSAGE);

        return;

    }

    this.dispose();

}

/**
 * @param args the command line arguments
 */
public static void main(String args[]) {

    /* Set the Nimbus look and feel */

    //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">

    /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.

    * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html

    */

```

```

try {
    for (javax.swing.UIManager.LookAndFeelInfo info :
        javax.swing.UIManager.getInstalledLookAndFeels()) {
        if ("Nimbus".equals(info.getName())) {
            javax.swing.UIManager.setLookAndFeel(info.getClassName());
            break;
        }
    }
} catch (ReflectiveOperationException |
        javax.swing.UnsupportedLookAndFeelException ex) {
    logger.log(java.util.logging.Level.SEVERE, null, ex);
}
//</editor-fold>

```

```

/* Create and display the form */

try {
    for (javax.swing.UIManager.LookAndFeelInfo info :
        javax.swing.UIManager.getInstalledLookAndFeels()) {
        if ("Nimbus".equals(info.getName())) {
            javax.swing.UIManager.setLookAndFeel(info.getClassName());
            break;
        }
    }
} catch (Exception ex) {
    logger.log(java.util.logging.Level.SEVERE, null, ex);
}

java.awt.EventQueue.invokeLater(() -> {

```

```
        AppContext context = new AppContext();
        new LoginFrame(context).setVisible(true);
    });
}
```

```
// Variables declaration - do not modify

private javax.swing.JButton btnLogin;
private javax.swing.JTextField idText;
private javax.swing.JLabel jLabel4;
private javax.swing.JScrollBar jScrollBar1;
private javax.swing.JLabel lblId;
private javax.swing.JLabel lblPassword;
private javax.swing.JLabel lblTitulo;
private javax.swing.JLabel mensajeLbl;
private javax.swing.JPasswordField passwordText;

// End of variables declaration
}

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to
change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to edit
this template
 */

package interfaz;

import java.time.LocalDateTime;
import java.util.ArrayList;
```

```
import javax.swing.*;

import modelo.*;

import servicio.*;

/**
 *
 * @author sam10
 */
public class AdminFrame extends javax.swing.JFrame {

    private AppContext context;

    private Admin admin;

    private AsistenciaServicio asistenciaServicio;

    private UsuarioServicio usuarioServicio;

    private CursoServicio cursoServicio;

    private ConstanciaServicio constanciaServicio;

    private static final java.util.logging.Logger logger =
    java.util.logging.Logger.getLogger(AdminFrame.class.getName());

    /**
     * Creates new form AdminFrame
     */
    public AdminFrame(AppContext context, Admin admin) {

        this.context = context;

        this.admin = admin;
```

```

this.asistenciaServicio = context.getAsistenciaServicio();
this.usuarioServicio = context.getUsuarioServicio();
this.cursoServicio = context.getCursoServicio();
this.constanciaServicio = context.getConstanciaServicio();

initComponents();

setLocationRelativeTo(null);

cargarEstuYCursos();
menuVerDatos.addActionListener(e -> {
    String mensaje = "Nombre: " + admin.getNombre()
        + "\nID: " + admin.getId();
    JOptionPane.showMessageDialog(this, mensaje,
        "Datos personales", JOptionPane.INFORMATION_MESSAGE);
});
}

/**
 * This method is called from within the constructor to initialize the form.
 * WARNING: Do NOT modify this code. The content of this method is always
 * regenerated by the Form Editor.
 */

@SuppressWarnings("unchecked")
// <editor-fold defaultstate="collapsed" desc="Generated Code">
private void initComponents() {

```

```
jTabbedPane1 = new javax.swing.JTabbedPane();
tabbedPaneAdmin = new javax.swing.JTabbedPane();
jPanel1 = new javax.swing.JPanel();
btnRegistrarEntrada = new javax.swing.JButton();
btnRegistrarSalida = new javax.swing.JButton();
btnVerReporte = new javax.swing.JButton();
jScrollPane1 = new javax.swing.JScrollPane();
txtReporteAsis = new javax.swing.JTextArea();
jPanel2 = new javax.swing.JPanel();
jScrollPane2 = new javax.swing.JScrollPane();
listEstudiantes = new javax.swing.JList<>();
jScrollPane3 = new javax.swing.JScrollPane();
listCursos = new javax.swing.JList<>();
btnAsignarEstCurso = new javax.swing.JButton();
jPanel3 = new javax.swing.JPanel();
btnAgregarCurso = new javax.swing.JButton();
btonAgregarUsuario = new javax.swing.JButton();
jMenuBar1 = new javax.swing.JMenuBar();
jMenu1 = new javax.swing.JMenu();
menuVerDatos = new javax.swing.JMenuItem();
menuCerrarSesion = new javax.swing.JMenuItem();

setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

tabbedPaneAdmin.setName(""); // NOI18N
```

```
btnRegistrarEntrada.setText("Registrar asistencia");  
btnRegistrarEntrada.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnRegistrarEntradaActionPerformed(evt);  
    }  
});
```

```
btnRegistrarSalida.setText("Registrar salida");  
btnRegistrarSalida.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnRegistrarSalidaActionPerformed(evt);  
    }  
});
```

```
btnVerReporte.setText("Ver reporte");  
btnVerReporte.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnVerReporteActionPerformed(evt);  
    }  
});
```

```
txtReporteAsis.setEditable(false);  
txtReporteAsis.setColumns(20);  
txtReporteAsis.setRows(5);  
jScrollPane1.setViewportView(txtReporteAsis);
```

```

    javax.swing.GroupLayout jPanel1Layout = new javax.swing.GroupLayout(jPanel1);
    jPanel1.setLayout(jPanel1Layout);
    jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(jPanel1Layout.createSequentialGroup()

        .addGap(25, 25, 25)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.L
EADING)

    .addComponent(btnRegistrarEntrada)

    .addComponent(btnVerReporte)

    .addComponent(btnRegistrarSalida))

    .addGap(18, 18, 18)

    .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 221,
Short.MAX_VALUE)

    .addContainerGap())
);
jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

    .addGroup(jPanel1Layout.createSequentialGroup()

        .addGap(57, 57, 57)

        .addComponent(btnRegistrarEntrada)

        .addGap(29, 29, 29)

        .addComponent(btnRegistrarSalida)

        .addGap(30, 30, 30)

```



```
.addComponent(btnVerReporte)

.addContainerGap(86, Short.MAX_VALUE))

.addComponent(jScrollPane1, javax.swing.GroupLayout.Alignment.TRAILING)

);
```

```
tabbedPaneAdmin.addTab("Asistencia", jPanel1);
```

```
listEstudiantes.setModel(new javax.swing.AbstractListModel<String>() {

    String[] strings = { "Item 1", "Item 2", "Item 3", "Item 4", "Item 5" };

    public int getSize() { return strings.length; }

    public String getElementAt(int i) { return strings[i]; }

});

jScrollPane2.setViewportViewView(listEstudiantes);
```

```
listCursos.setModel(new javax.swing.AbstractListModel<String>() {

    String[] strings = { "Item 1", "Item 2", "Item 3", "Item 4", "Item 5" };

    public int getSize() { return strings.length; }

    public String getElementAt(int i) { return strings[i]; }

});

jScrollPane3.setViewportViewView(listCursos);
```

```
btnAsignarEstCurso.setText("Asignar estudiante a curso");

btnAsignarEstCurso.addActionListener(new java.awt.event.ActionListener() {

    public void actionPerformed(java.awt.event.ActionEvent evt) {

        btnAsignarEstCursoActionPerformed(evt);

    }

});
```

```

});

javax.swing.GroupLayout jPanel2Layout = new javax.swing.GroupLayout(jPanel2);
jPanel2.setLayout(jPanel2Layout);
jPanel2Layout.setHorizontalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(jPanel2Layout.createSequentialGroup()
        .addContainerGap()
        .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(jScrollPane2, javax.swing.GroupLayout.PREFERRED_SIZE, 97,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(49, 49, 49)
            .addComponent(jScrollPane3, javax.swing.GroupLayout.PREFERRED_SIZE, 108,
                javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGap(93, 93, 93)
                .addComponent(btnAsignarEstCurso)))
        .addGap(74, 74, Short.MAX_VALUE))
    );

jPanel2Layout.setVerticalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(jPanel2Layout.createSequentialGroup()
        .addContainerGap()
        .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(jScrollPane2, javax.swing.GroupLayout.PREFERRED_SIZE, 97,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addGap(49, 49, 49)
            .addComponent(jScrollPane3, javax.swing.GroupLayout.PREFERRED_SIZE, 108,
                javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGroup(jPanel2Layout.createSequentialGroup()
                .addGap(93, 93, 93)
                .addComponent(btnAsignarEstCurso)))
        .addGap(74, 74, Short.MAX_VALUE))
    );

```

```

        .addGap(26, 26, 26)

        .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.TRAILING))

            .addComponent(jScrollPane3, javax.swing.GroupLayout.PREFERRED_SIZE,
                javax.swing.GroupLayout.DEFAULT_SIZE,
                javax.swing.GroupLayout.PREFERRED_SIZE)

            .addComponent(jScrollPane2, javax.swing.GroupLayout.PREFERRED_SIZE,
                javax.swing.GroupLayout.DEFAULT_SIZE,
                javax.swing.GroupLayout.PREFERRED_SIZE))

        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED,
            javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)

        .addComponent(btnAsignarEstCurso)

        .addGap(30, 30, 30))

    );

```

```

tabbedPaneAdmin.addTab("Estudiantes", jPanel2);

```

```

btnAgregarCurso.setText("Agregar Curso");

btnAgregarCurso.addActionListener(new java.awt.event.ActionListener() {

    public void actionPerformed(java.awt.event.ActionEvent evt) {

        btnAgregarCursoActionPerformed(evt);

    }

});

```

```

btonAgregarUsuario.setText("Agregar usuario");

btonAgregarUsuario.addActionListener(new java.awt.event.ActionListener() {

    public void actionPerformed(java.awt.event.ActionEvent evt) {

```

```

        botonAgregarUsuarioActionPerformed(evt);
    }
});

javax.swing.GroupLayout jPanel3Layout = new javax.swing.GroupLayout(jPanel3);
jPanel3.setLayout(jPanel3Layout);
jPanel3Layout.setHorizontalGroup(

jPanel3Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(jPanel3Layout.createSequentialGroup()
        .addGap(47, 47, 47)

        .addGroup(jPanel3Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(btnAgregarCurso)
            .addGroup(jPanel3Layout.createSequentialGroup()
                .addComponent(btonAgregarUsuario,
                    javax.swing.GroupLayout.PREFERRED_SIZE, 126,
                    javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGap(229, Short.MAX_VALUE))
        );

jPanel3Layout.setVerticalGroup(

jPanel3Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(jPanel3Layout.createSequentialGroup()
        .addGap(56, 56, 56)
        .addComponent(btnAgregarCurso)
        .addGap(52, 52, 52)
        .addComponent(btonAgregarUsuario)
    )
);

```

```

        .addContainerGap(117, Short.MAX_VALUE))
    );

    tabbedPaneAdmin.addTab("Administracion", jPanel3);

    jMenu1.setText("Cuenta");

    menuVerDatos.setText("Datos");
    jMenu1.add(menuVerDatos);

    menuCerrarSesion.setText("Cerrar sesión");
    menuCerrarSesion.addActionListener(new java.awt.event.ActionListener() {
        public void actionPerformed(java.awt.event.ActionEvent evt) {
            menuCerrarSesionActionPerformed(evt);
        }
    });
    jMenu1.add(menuCerrarSesion);

    jMenuBar1.add(jMenu1);

    setJMenuBar(jMenuBar1);

    javax.swing.GroupLayout layout = new
    javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(

```

```
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(tabbedPaneAdmin)
    );

    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(tabbedPaneAdmin,
                javax.swing.GroupLayout.Alignment.TRAILING)
    );

    pack();
} // </editor-fold>
```

```
private void cargarEstuYCursos() {
    cargarListaEstudiantes();
    cargarListaCursos();
}
```

```
private void cargarListaEstudiantes() {
    DefaultListModel<String> modeloEst = new DefaultListModel<>();
    for (Usuario u : usuarioServicio.getUsuarios()) {
        if (u instanceof Estudiante) {
            modeloEst.addElement(u.getId() + " - " + u.getNombre());
        }
    }
    listEstudiantes.setModel(modeloEst);
}
```

```

private void cargarListaCursos() {

    DefaultListModel<String> modeloCur = new DefaultListModel<>();

    for (Curso c : cursoServicio.getCursos()) {

        modeloCur.addElement(c.getNombre());

    }

    listCursos.setModel(modeloCur);

}

```

```

private void actualizarReporteAsistencia() {

    ArrayList<Asistencia> reporte = asistenciaServicio.obtenerReporteAsis(admin);

    StringBuilder sb = new StringBuilder();

    if (reporte.isEmpty()) {

        sb.append("No hay asistencias registradas.\n");

    }

    else {

        for (Asistencia a : reporte) {

            String textoSalida = (a.getFechaHoraSalida() == null)

                ? "Sin registrar"

                : a.getFechaHoraSalida().toString();

            sb.append("Usuario: ").append(a.getUsuario().getId()).append("\n");

            sb.append("Entrada: ").append(a.getFechaHoraEntrada()).append("\n");

            sb.append("Salida: ").append(textoSalida).append("\n");

            sb.append("-----\n");

        }

    }

}

```

```

    }
}

txtReporteAsis.setText(sb.toString());
}

private void menuCerrarSesionActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    new LoginFrame(context).setVisible(true);
    dispose();
}

private void btnRegistrarEntradaActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    Asistencia a = new Asistencia(admin, LocalDateTime.now());
    asistenciaServicio.registrarAsistencia(a);
    JOptionPane.showMessageDialog(this, "Asistencia registrada correctamente");
    actualizarReporteAsistencia();
}

private void btnRegistrarSalidaActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    boolean ok = asistenciaServicio.registrarSalida(admin, LocalDateTime.now());
    if (ok) {
        JOptionPane.showMessageDialog(this, "Salida registrada correctamente");
    } else {

```



```

        JOptionPane.showMessageDialog(this,

            "No se encontró una asistencia sin salida",

            "Aviso",

            JOptionPane.WARNING_MESSAGE);
    }

    actualizarReporteAsistencia();
}

private void btnVerReporteActionPerformed(java.awt.event.ActionEvent evt) {

    // TODO add your handling code here:

    actualizarReporteAsistencia();
}

private void btnAsignarEstCursoActionPerformed(java.awt.event.ActionEvent evt) {

    // TODO add your handling code here:

    int idxEst = listEstudiantes.getSelectedIndex();

    int idxCur = listCursos.getSelectedIndex();

    if (idxEst == -1 || idxCur == -1) {

        JOptionPane.showMessageDialog(this,

            "Selecciona un estudiante y un curso",

            "Aviso",

            JOptionPane.WARNING_MESSAGE);

        return;
    }
}

```

```

    Estudiante est = obtenerEstudiantePorIndice(idxEst);
    Curso curso = cursoServicio.getCursos().get(idxCur);

    cursoServicio.asignarEstudiante(curso, est);
    JOptionPane.showMessageDialog(this,
        "Estudiante " + est.getId() + " asignado a " + curso.getNombre());
}

private void btnAgregarCursoActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:

    String nombreCur = JOptionPane.showInputDialog(this, "Nombre del curso:");
    if (nombreCur == null || nombreCur.isBlank()) return;

    Curso curso = new Curso(nombreCur);
    cursoServicio.agregarCurso(curso);

    String folio = "CONS" + curso.getNombre();
    ConstanciaCurso constancia = new ConstanciaCurso(folio);
    constanciaServicio.registrarConstancia(curso, constancia);

    JOptionPane.showMessageDialog(this,
        "Curso " + curso.getNombre() + " agregado.\nFolio base: " + folio);

    cargarListaCursos();
}

```

```

private void botonAgregarUsuarioActionPerformed(java.awt.event.ActionEvent evt) {

    // TODO add your handling code here:

    String[] opciones = {"Estudiante", "Administrador"};

    int tipo = JOptionPane.showOptionDialog(this,

        "Tipo de usuario:",

        "Agregar usuario",

        JOptionPane.DEFAULT_OPTION,

        JOptionPane.QUESTION_MESSAGE,

        null,

        opciones,

        opciones[0]);

    if (tipo == JOptionPane.CLOSED_OPTION) return;

    String nombre = JOptionPane.showInputDialog(this, "Nombre:");

    if (nombre == null || nombre.isBlank()) return;

    String pass = JOptionPane.showInputDialog(this, "Contraseña:");

    if (pass == null || pass.isBlank()) return;

    if (tipo == 0) { // Estudiante

        Estudiante est = new Estudiante(nombre, pass);

        usuarioServicio.agregarEstudiante(est);

        JOptionPane.showMessageDialog(this,

            "Estudiante agregado con ID: " + est.getId());

    }

    else { // Admin

```

```

        Admin admin1 = new Admin(nombre, pass);
        usuarioServicio.agregarAdmin(admin1);
        JOptionPane.showMessageDialog(this,
            "Administrador agregado con ID: " + admin1.getId());
    }
    cargarListaEstudiantes();
}

private Estudiante obtenerEstudiantePorIndice(int idxEst) {
    int contador = 0;
    for (Usuario u : usuarioServicio.getUsuarios()) {
        if (u instanceof Estudiante est) {
            if (contador == idxEst) {
                return est;
            }
            contador++;
        }
    }
    return null; // no debería pasar
}

```

```
/**
```

```
 * @param args the command line arguments
```

```
 */
```

```
public static void main(String args[]) {
```

```

/* Set the Nimbus look and feel */

//<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">

/* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.

* For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html

*/

try {

    for (javax.swing.UIManager.LookAndFeelInfo info :
        javax.swing.UIManager.getInstalledLookAndFeels()) {

        if ("Nimbus".equals(info.getName())) {

            javax.swing.UIManager.setLookAndFeel(info.getClassName());

            break;

        }

    }

} catch (ReflectiveOperationException |
        javax.swing.UnsupportedLookAndFeelException ex) {

    logger.log(java.util.logging.Level.SEVERE, null, ex);

}

//</editor-fold>

/* Create and display the form */

}

// Variables declaration - do not modify

private javax.swing.JButton btnAgregaCurso;

private javax.swing.JButton btnAsignarEstCurso;

```

```

private javax.swing.JButton btnRegistrarEntrada;
private javax.swing.JButton btnRegistrarSalida;
private javax.swing.JButton btnVerReporte;
private javax.swing.JButton botonAgregarUsuario;
private javax.swing.JMenu jMenu1;
private javax.swing.JMenuBar jMenuBar1;
private javax.swing.JPanel jPanel1;
private javax.swing.JPanel jPanel2;
private javax.swing.JPanel jPanel3;
private javax.swing.JScrollPane jScrollPane1;
private javax.swing.JScrollPane jScrollPane2;
private javax.swing.JScrollPane jScrollPane3;
private javax.swing.JTabbedPane jTabbedPane1;
private javax.swing.JList<String> listCursos;
private javax.swing.JList<String> listEstudiantes;
private javax.swing.JMenuItem menuCerrarSesion;
private javax.swing.JMenuItem menuVerDatos;
private javax.swing.JTabbedPane tabbedPaneAdmin;
private javax.swing.JTextArea txtReporteAsis;

// End of variables declaration
}

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to
change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/GUIForms/JFrame.java to edit
this template
 */

```

```
package interfaz;
```

```
import modelo.Asistencia;
```

```
import modeloCurso;
```

```
import modelo.Estudiante;
```

```
import modelo.Usuario;
```

```
import modelo.Asistencia;
```

```
import servicio.AsistenciaServicio;
```

```
import servicio.ConstanciaServicio;
```

```
import java.time.LocalDateTime;
```

```
import java.util.ArrayList;
```

```
import javax.swing.JOptionPane;
```

```
/**
```

```
 *
```

```
 * @author sam10
```

```
 */
```

```
public class EstudianteFrame extends javax.swing.JFrame {
```

```
    private ApplicationContext context;
```

```
    private Estudiante estudiante;
```

```
    private AsistenciaServicio asistenciaServicio;
```

```
    private ConstanciaServicio constanciaServicio;
```

```

private static final java.util.logging.Logger logger =
java.util.logging.Logger.getLogger(EstudianteFrame.class.getName());

/**
 * Creates new form EstudianteFrame
 */
public EstudianteFrame(AppContext context, Estudiante estudiante) {

    this.context = context;

    this.estudiante = estudiante;

    this.asistenciaServicio = context.getAsistenciaServicio();
    this.constanciaServicio = context.getConstanciaServicio();
    initComponents();
    setLocationRelativeTo(null);
    setTitle("Panel estudiante -" + estudiante.getNombre());
    cargarCursosEnCombo();

    menuVerDatos.addActionListener(e -> {

        String mensaje = "Nombre: " + estudiante.getNombre()
            + "\nID: " + estudiante.getId();

        JOptionPane.showMessageDialog(this, mensaje,
            "Datos personales", JOptionPane.INFORMATION_MESSAGE);

    });
}

/**
 * This method is called from within the constructor to initialize the form.

```


* WARNING: Do NOT modify this code. The content of this method is always

* regenerated by the Form Editor.

*/

@SuppressWarnings("unchecked")

// <editor-fold defaultstate="collapsed" desc="Generated Code">

private void initComponents() {

jTabbedPane1 = new javax.swing.JTabbedPane();

jPanel1 = new javax.swing.JPanel();

btnRegistrarEntrada = new javax.swing.JButton();

btnRegistrarSalida = new javax.swing.JButton();

btnVerReporte = new javax.swing.JButton();

jScrollPane1 = new javax.swing.JScrollPane();

txtReporteAsistencia = new javax.swing.JTextArea();

jPanel2 = new javax.swing.JPanel();

jLabel1 = new javax.swing.JLabel();

comboCursos = new javax.swing.JComboBox<>();

jLabel2 = new javax.swing.JLabel();

txtFolio = new javax.swing.JTextField();

btnValidarFolio = new javax.swing.JButton();

lblResultadoFolio = new javax.swing.JLabel();

jMenuBar1 = new javax.swing.JMenuBar();

jMenu1 = new javax.swing.JMenu();

jMenuItem2 = new javax.swing.JMenuItem();

menuVerDatos = new javax.swing.JMenuItem();

```
setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
```

```
btnRegistrarEntrada.setText("Registrar entrada");
```

```
btnRegistrarEntrada.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnRegistrarEntradaActionPerformed(evt);  
    }  
});
```

```
btnRegistrarSalida.setText("Registrar salida");
```

```
btnRegistrarSalida.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnRegistrarSalidaActionPerformed(evt);  
    }  
});
```

```
btnVerReporte.setText("Ver reporte");
```

```
btnVerReporte.addActionListener(new java.awt.event.ActionListener() {  
    public void actionPerformed(java.awt.event.ActionEvent evt) {  
        btnVerReporteActionPerformed(evt);  
    }  
});
```

```
txtReporteAsistencia.setColumns(20);
```

```
txtReporteAsistencia.setRows(5);
```

```
jScrollPane1.setViewportView(txtReporteAsistencia);
```

```

    javax.swing.GroupLayout jPanel1Layout = new javax.swing.GroupLayout(jPanel1);
    jPanel1.setLayout(jPanel1Layout);
    jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(jPanel1Layout.createSequentialGroup()
        .addGap(37, 37, 37)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addComponent(btnRegistrarEntrada)
    .addComponent(btnRegistrarSalida)
    .addGroup(jPanel1Layout.createSequentialGroup()
        .addGap(8, 8, 8)
        .addComponent(btnVerReporte)))
    .addGap(18, 18, 18)
    .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE,
247, javax.swing.GroupLayout.PREFERRED_SIZE))
    );
    jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
    .addGroup(jPanel1Layout.createSequentialGroup()
        .addGap(38, 38, 38)
        .addComponent(btnRegistrarEntrada)
        .addGap(45, 45, 45)
        .addComponent(btnRegistrarSalida)

```

```

        .addGap(45, 45, 45)

        .addComponent(btnVerReporte)

        .addContainerGap(45, Short.MAX_VALUE))

        .addComponent(jScrollPane1)

    );

    jTabbedPane1.addTab("Asistencia", jPanel1);

    jLabel1.setText("Selecciona un curso:");

    comboCursos.setModel(new javax.swing.DefaultComboBoxModel<>(new String[]
{ "Item 1", "Item 2", "Item 3", "Item 4" }));

    jLabel2.setText("Folio:");

    btnValidarFolio.setText("Validar folio");

    btnValidarFolio.addActionListener(new java.awt.event.ActionListener() {

        public void actionPerformed(java.awt.event.ActionEvent evt) {

            btnValidarFolioActionPerformed(evt);

        }

    });

    javax.swing.GroupLayout jPanel2Layout = new javax.swing.GroupLayout(jPanel2);
    jPanel2.setLayout(jPanel2Layout);
    jPanel2Layout.setHorizontalGroup(

        jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

    );

```

```

        .addGroup(jPanel2Layout.createSequentialGroup())

        .addGap(95, 95, 95)

    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

        .addComponent(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE, 37,
        javax.swing.GroupLayout.PREFERRED_SIZE)

        .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 115,
        javax.swing.GroupLayout.PREFERRED_SIZE))

        .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED,
        51, Short.MAX_VALUE)

    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

        .addComponent(comboCursos,
        javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
        javax.swing.GroupLayout.PREFERRED_SIZE)

        .addComponent(txtFolio, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
        javax.swing.GroupLayout.PREFERRED_SIZE))

        .addGap(90, 90, 90))

    .addGroup(jPanel2Layout.createSequentialGroup())

        .addGap(165, 165, 165)

    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

        .addComponent(lblResultadoFolio,
        javax.swing.GroupLayout.PREFERRED_SIZE, 111,
        javax.swing.GroupLayout.PREFERRED_SIZE)

        .addComponent(btnValidarFolio))

        .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
        Short.MAX_VALUE))

```

```

);

jPanel2Layout.setVerticalGroup(

jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)

    .addGroup(jPanel2Layout.createSequentialGroup()

        .addGap(38, 38, 38)

    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)

        .addComponent(jLabel1)

        .addComponent(comboCursos,
javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))

        .addGap(44, 44, 44)

    .addGroup(jPanel2Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)

        .addComponent(jLabel2)

        .addComponent(txtFolio, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))

        .addGap(30, 30, 30)

        .addComponent(btnValidarFolio)

    .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)

        .addComponent(lblResultadoFolio,
javax.swing.GroupLayout.PREFERRED_SIZE, 18,
javax.swing.GroupLayout.PREFERRED_SIZE)

        .addContainerGap(33, Short.MAX_VALUE)

    );

```

```
jTabbedPane1.addTab("Cursos", jPanel2);
```

```
jMenu1.setText("Cuenta");
```

```
jMenuItem2.setText("Datos");
```

```
jMenu1.add(jMenuItem2);
```

```
menuVerDatos.setText("Cerrar sesión");
```

```
menuVerDatos.addActionListener(new java.awt.event.ActionListener() {
```

```
    public void actionPerformed(java.awt.event.ActionEvent evt) {
```

```
        menuVerDatosActionPerformed(evt);
```

```
    }
```

```
});
```

```
jMenu1.add(menuVerDatos);
```

```
jMenuBar1.add(jMenu1);
```

```
setJMenuBar(jMenuBar1);
```

```
javax.swing.GroupLayout layout = new  
javax.swing.GroupLayout(getContentPane());
```

```
getContentPane().setLayout(layout);
```

```
layout.setHorizontalGroup(
```

```
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
        .addComponent(jTabbedPane1, javax.swing.GroupLayout.Alignment.TRAILING)
```

```
);  
  
layout.setVerticalGroup(  
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)  
    .addComponent(jTabbedPane1, javax.swing.GroupLayout.Alignment.TRAILING)  
);  
  
pack();  
} // </editor-fold>
```

```
private void cargarCursosEnCombo() {  
    comboCursos.removeAllItems();  
  
    if (estudiante.getCursos() == null || estudiante.getCursos().isEmpty()) {  
        comboCursos.addItem("Sin cursos");  
        comboCursos.setEnabled(false);  
    }  
  
    javax.swing.DefaultComboBoxModel<Curso> modelo = new  
    javax.swing.DefaultComboBoxModel<>();  
  
    for (Curso c : estudiante.getCursos()) {  
        comboCursos.addItem(c.getNombre());  
    }  
    comboCursos.setEnabled(true);  
}
```



```

private void actualizarReporteAsistencia() {

    ArrayList<Asistencia> reporte =
asistenciaServicio.obtenerReporteAsis(estudiante);

    StringBuilder sb = new StringBuilder();

    if (reporte.isEmpty()) {

        sb.append("No hay asistencias registradas.\n");

    }
    else {

        for (Asistencia a : reporte) {

            String textoSalida = (a.getFechaHoraSalida() == null)

                ? "Sin registrar"

                : a.getFechaHoraSalida().toString();

            sb.append("Usuario: ").append(a.getUsuario().getId()).append("\n");

            sb.append("Entrada: ").append(a.getFechaHoraEntrada()).append("\n");

            sb.append("Salida: ").append(textoSalida).append("\n");

            sb.append("-----\n");

        }

    }

    txtReporteAsistencia.setText(sb.toString());

}

private void btnRegistrarEntradaActionPerformed(java.awt.event.ActionEvent evt) {

    // TODO add your handling code here:

```

```

Asistencia a = new Asistencia(estudiante, LocalDateTime.now());
asistenciaServicio.registrarAsistencia(a);
JOptionPane.showMessageDialog(this, "Asistencia registrada correctamente");
actualizarReporteAsistencia();
}

private void btnRegistrarSalidaActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    boolean ok = asistenciaServicio.registrarSalida(estudiante, LocalDateTime.now());
    if (ok) {
        JOptionPane.showMessageDialog(this, "Salida registrada correctamente");
    } else {
        JOptionPane.showMessageDialog(this,
            "No se encontró una asistencia sin salida",
            "Aviso",
            JOptionPane.WARNING_MESSAGE);
    }
    actualizarReporteAsistencia();
}

private void btnVerReporteActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    actualizarReporteAsistencia();
}

private void btnValidarFolioActionPerformed(java.awt.event.ActionEvent evt) {

```

```
// TODO add your handling code here:

Curso cursoSeleccionado = null;

String folio = txtFolio.getText().trim();


String nombreCursoSeleccionado = (String) comboCursos.getSelectedItemId();


if (nombreCursoSeleccionado == null || nombreCursoSeleccionado.equals("Sin
cursos")) {

    JOptionPane.showMessageDialog(this, "Selecciona un curso valido",
        "Aviso", JOptionPane.WARNING_MESSAGE);
}


for (Curso c : estudiante.getCursos()) {

    if (c.getNombre().equals(nombreCursoSeleccionado)) {

        cursoSeleccionado = c;

        break;

    }

}


if (cursoSeleccionado == null) {

    JOptionPane.showMessageDialog(this,
        "No se encontro el curso seleccionado",
        "Error", JOptionPane.ERROR_MESSAGE);

    return;

}
```

```
if (folio.isEmpty()) {  
    JOptionPane.showMessageDialog(this, "Ingresa un folio"  
        , "Aviso", JOptionPane.WARNING_MESSAGE);  
    return;  
}
```

```
boolean esValido = constanciaServicio.validarFolioDeCurso(cursoSeleccionado,  
folio);
```

```
if (esValido) {  
    lblResultadoFolio.setText("Folio válido");  
    lblResultadoFolio.setForeground(new java.awt.Color(0, 128, 0));  
    JOptionPane.showMessageDialog(this, "Folio válido");  
}  
else {  
    lblResultadoFolio.setText("Folio inválido");  
    lblResultadoFolio.setForeground(java.awt.Color.RED);  
    JOptionPane.showMessageDialog(this, "Folio inválido");  
}  
}
```

```
private void menuVerDatosActionPerformed(java.awt.event.ActionEvent evt) {  
    // TODO add your handling code here:  
    new LoginFrame(context).setVisible(true);  
    dispose();  
}
```

```

}

/**
 * @param args the command line arguments
 */
public static void main(String args[]) {

    /* Set the Nimbus look and feel */

    //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">

    /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.

    * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
    */

    try {

        for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {

            if ("Nimbus".equals(info.getName())) {

                javax.swing.UIManager.setLookAndFeel(info.getClassName());

                break;

            }

        }

    } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException ex) {

        logger.log(java.util.logging.Level.SEVERE, null, ex);

    }

    //</editor-fold>

```

```
}
```

```
// Variables declaration - do not modify
```

```
private javax.swing.JButton btnRegistrarEntrada;
```

```
private javax.swing.JButton btnRegistrarSalida;
```

```
private javax.swing.JButton btnValidarFolio;
```

```
private javax.swing.JButton btnVerReporte;
```

```
private javax.swing.JComboBox<String> comboCursos;
```

```
private javax.swing.JLabel jLabel1;
```

```
private javax.swing.JLabel jLabel2;
```

```
private javax.swing.JMenu jMenu1;
```

```
private javax.swing.JMenuBar jMenuBar1;
```

```
private javax.swing.JMenuItem jMenuItem2;
```

```
private javax.swing.JPanel jPanel1;
```

```
private javax.swing.JPanel jPanel2;
```

```
private javax.swing.JScrollPane jScrollPane1;
```

```
private javax.swing.JTabbedPane jTabbedPane1;
```

```
private javax.swing.JLabel lblResultadoFolio;
```

```
private javax.swing.JMenuItem menuVerDatos;
```

```
private javax.swing.JTextField txtFolio;
```

```
private javax.swing.JTextArea txtReporteAsistencia;
```

```
// End of variables declaration
```

```
}
```

```
package interfaz;
```

```
import java.util.ArrayList;

import modelo.*;

import servicio.*;

/**
 *
 * @author sam10
 */
public class AppContext {

    private ArrayList<Usuario> usuarios;

    private ArrayList<Curso> cursos;

    private UsuarioServicio usuarioServicio;

    private CursoServicio cursoServicio;

    private AsistenciaServicio asistenciaServicio;

    private ConstanciaServicio constanciaServicio;


    public AppContext() {

        usuarios = new ArrayList<>();

        cursos = new ArrayList<>();

        usuarioServicio = new UsuarioServicio(usuarios);

        cursoServicio = new CursoServicio(cursos);

        asistenciaServicio = new AsistenciaServicio();

        constanciaServicio = new ConstanciaServicio();


        Admin admi1 = new Admin("Abigail", "admin1");
```

```
Estudiante est1 = new Estudiante ("Maria", "estu1");  
Estudiante est2 = new Estudiante ("Ruben", "estu4");
```

```
usuarios.add(admi1);  
usuarios.add(est1);  
usuarios.add(est2);  
}
```

```
public Usuario iniciarSesion(String id, String contrasenia) {  
    for (Usuario u : usuarios) {  
        if (u.getId().equalsIgnoreCase(id) &&  
            u.verificarContrasena(contrasenia)) {  
            return u;  
        }  
    }  
    return null;  
}
```

```
public UsuarioServicio getUsuarioServicio() {  
    return usuarioServicio;  
}
```

```
public CursoServicio getCursoServicio() {  
    return cursoServicio;  
}
```



```
public AsistenciaServicio getAsistenciaServicio() {  
    return asistenciaServicio;  
}
```

```
public ConstanciaServicio getConstanciaServicio() {  
    return constanciaServicio;  
}
```

```
}
```

```
package modelo;
```

```
/**
```

```
 *
```

```
 * @author sam10
```

```
 */
```

```
public class Admin extends Usuario {  
    private static int cont = 1;  
    public Admin (String nombre, String contrasenia){  
        super(nombre, contrasenia);  
        generarId();  
    }  
}
```

```
@Override
```

```
public void generarId(){
```

```
        this.id= "AD" + cont;

        cont++;
    }
}

package modelo;

import java.time.LocalDateTime;

/**
 *
 * @author sam10
 */
public class Asistencia {

    private Usuario usuario;

    private LocalDateTime fechaHoraEntrada;

    private LocalDateTime fechaHoraSalida;


    public Asistencia(Usuario usuario, LocalDateTime fechaHoraEntrada){

        this.usuario = usuario;

        this.fechaHoraEntrada = fechaHoraEntrada;

    }


    public void registrarSalida(LocalDateTime fechaHoraSalida){

        this.fechaHoraSalida = fechaHoraSalida;

    }

}
```

```

    public Usuario getUsuario(){
        return usuario;
    }

    public LocalDateTime getFechaHoraEntrada() {
        return fechaHoraEntrada;
    }

    public LocalDateTime getFechaHoraSalida(){
        return fechaHoraSalida;
    }

}

package modelo;

/**
 *
 * @author sam10
 */
public class ConstanciaCurso {
    private String folio;

    public ConstanciaCurso(String folio){
        this.folio = folio;
    }

}

```

```

        public String getFolio(){
            return folio;
        }

    }

package modelo;

import java.util.ArrayList;
import java.util.HashMap;

/**
 *
 * @author sam10
 */
public class Curso {
    private String nombre;
    private ArrayList<Estudiante> estudiantes;
    private HashMap<String, ConstanciaCurso> foliosValidos;

    public Curso(String nombre){
        this.nombre = nombre;
        this.estudiantes = new ArrayList<>();
        this.foliosValidos = new HashMap<>();
    }

```

```

public void agregarEstudiante(Estudiante e){
    if(!estudiantes.contains(e)){
        estudiantes.add(e);
        e.agregarCurso(this);
    }
}

public void registrarFolioValido(ConstanciaCurso constancia){
    foliosValidos.put(constancia.getFolio(), constancia);
}

public boolean esFolioValido(String folio) {
    return foliosValidos.containsKey(folio);
}

public String getNombre(){
    return nombre;
}

public ArrayList<Estudiante> getEstudiantes() {
    return estudiantes;
}
}

package modelo;

import java.util.ArrayList;

```

```
/**
```

```
*
```

```
* @author sam10
```

```
*/
```

```
public class Estudiante extends Usuario {
```

```
    private static int cont = 1;
```

```
    private ArrayList<Curso> cursos;
```

```
    public Estudiante (String nombre, String contrasenia){
```

```
        super(nombre, contrasenia);
```

```
        this.cursos = new ArrayList<>();
```

```
        generarId();
```

```
    }
```

```
    @Override
```

```
    public void generarId(){
```

```
        this.id= "ES" + cont;
```

```
        cont++;
```

```
    }
```

```
    public void agregarCurso(Curso curso){
```

```
        if (!cursos.contains(curso)) {
```

```
            cursos.add(curso);
```

```
        }
```

```
}
```

```
public ArrayList<Curso> getCursos() {
```

```
    return cursos;
```

```
}
```

```
}
```

```
package modelo;
```

```
/**
```

```
 *
```

```
 * @author sam10
```

```
 */
```

```
public abstract class Usuario {
```

```
    protected String id;
```

```
    protected String nombre;
```

```
    protected String contrasenia;
```

```
    public Usuario(String nombre, String contrasenia){
```

```
        this.nombre = nombre;
```

```
        this.contrasenia = contrasenia;
```

```
    }
```

```
    public boolean verificarContrasena(String contrasenia){
```

```
        return this.contrasenia.equals(contrasenia);
```

```
    }
```

```

    public abstract void generarId();

    public String getId(){
        return id;
    }

    public String getNombre() {
        return nombre;
    }

}

package servicio;

import interfaces.IAsistenciaServicio;
import java.time.LocalDateTime;
import java.util.ArrayList;
import modelo.Asistencia;
import modelo.Usuario;

/**
 *
 * @author sam10
 */
public class AsistenciaServicio implements IAsistenciaServicio {
    private ArrayList<Asistencia> asistencias;

```



```
public AsistenciaServicio(){  
    this.asistencias = new ArrayList<>();  
}
```

@Override

```
public void registrarAsistencia(Asistencia asistencia){  
    asistencias.add(asistencia);  
}
```

@Override

```
public boolean registrarSalida(Usuario usuario,  
    LocalDateTime fechaHoraSalida){  
    for (int i = asistencias.size() - 1; i >= 0; i--){  
        Asistencia reg = asistencias.get(i);  
        if(reg.getUsuario().equals(usuario)  
            && reg.getFechaHoraSalida() == null){  
            reg.registrarSalida(fechaHoraSalida);  
            return true;  
        }  
    }  
    return false;  
}
```

@Override

```
public ArrayList<Asistencia> obtenerReporteAsis(Usuario usuario){  
    ArrayList<Asistencia> resultado = new ArrayList<>();
```

```
        for (Asistencia a : asistencias){  
            if(a.getUsuario().equals(usuario)){  
                resultado.add(a);  
            }  
        }  
        return resultado;  
    }  
}
```

```
package servicio;
```

```
import modelo.ConstanciaCurso;
```

```
import modelo.Curso;
```

```
/**
```

```
 *
```

```
 * @author sam10
```

```
 */
```

```
public class ConstanciaServicio {
```

```
    public void registrarConstancia(Curso curso, ConstanciaCurso constancia) {
```

```
        curso.registrarFolioValido(constancia);
```

```
    }
```

```
    public boolean validarFolioDeCurso(Curso curso, String folio){
```

```
        return curso.esFolioValido(folio);
```

```
    }
```

```
}  
  
package servicio;  
  
import java.util.ArrayList;  
import modeloCurso;  
import modelo.Estudiante;  
  
/**  
 *  
 * @author sam10  
 */  
public class CursoServicio {  
    private ArrayList<Curso> cursos;  
  
    public CursoServicio(ArrayList<Curso> cursos){  
        this.cursos = cursos;  
    }  
  
    public void agregarCurso(Curso curso){  
        cursos.add(curso);  
    }  
  
    public void asignarEstudiante(Curso curso, Estudiante estudiante){  
        curso.agregarEstudiante(estudiante);  
    }  
}
```

```

public Curso buscarCurso(String nombre) {
    for (Curso c : cursos) {
        if (c.getNombre().equalsIgnoreCase(nombre)) {
            return c;
        }
    }
    return null;
}

```

```

public ArrayList<Curso> getCursos() {
    return cursos;
}
}

```

```

package servicio;

```

```

import java.util.ArrayList;
import modelo.Admin;
import modelo.Estudiante;
import modelo.Usuario;

```

```

/**

```

```

 *

```

```

 * @author sam10

```

```

 */

```

```

public class UsuarioServicio {
    private ArrayList<Usuario> usuarios;

```

```
public UsuarioServicio(ArrayList<Usuario> usuarios){  
    this.usuarios = usuarios;  
}  
  
public void agregarEstudiante(Estudiante estudiante){  
    usuarios.add(estudiante);  
}  
  
public void agregarAdmin(Admin admin){  
    usuarios.add(admin);  
}  
  
public Usuario buscarId(String id) {  
    for (Usuario u : usuarios) {  
        if (u.getId().equalsIgnoreCase(id)) {  
            return u;  
        }  
    }  
    return null;  
}  
  
public ArrayList<Usuario> getUsuarios() {  
    return usuarios;  
}  
}
```